

Measurement Systems and Transducers

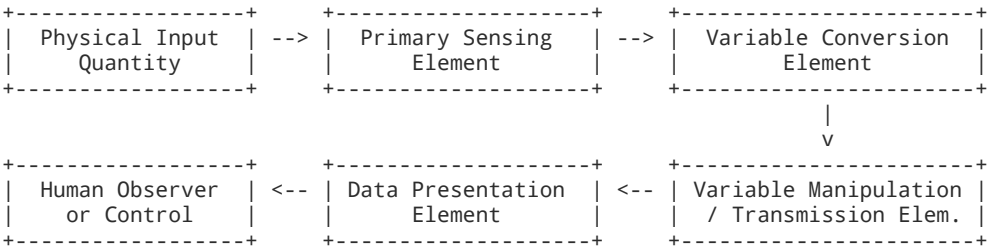
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User prompt: Unit-IV: Functional Elements of Instruments, Classification & Characteristics, Types of Errors, Sources of Error, Dynamic Characteristics, Active and Passive Transducers: Resistive Transducers, Thermistor, Strain Gauge, Thermocouple, Differential Output Transducers, LVDT and their Characteristics. Give proper extended explanation without missing anything give it to pdf

Response: Unit-IV: Measurement Systems, Transducers, and Instrumentation

1. Functional Elements of an Instrumentation System

An instrument or measurement system consists of several functional elements arranged in a series or parallel chain to measure a physical variable and convert it into a human-readable or machine-processable form.



1.1 Primary Sensing Element

- **Role:** Directly interfaces with the physical medium to detect the measurand (e.g., pressure, temperature, displacement).
- **Operation:** Absorbs a minute fraction of energy from the measurand and produces an output state dependent on that input.
- **Example:** A Bourdon tube sensing fluid pressure, a diaphragm deflecting under pressure, or a bimetallic strip bending due to temperature change.

1.2 Variable Conversion Element (Transducing Element)

- **Role:** Converts the output signal of the primary sensing element (often mechanical displacement or force) into a more usable physical form, typically an electrical parameter (voltage, current, resistance, capacitance).
- **Operation:** Acts without altering the informational content of the signal.
- **Example:** A linear variable differential transformer (LVDT) converting the mechanical displacement of a Bourdon tube into an AC voltage.

1.3 Variable Manipulation Element (Signal Conditioning)

- **Role:** Modifies the converted signal to make it suitable for transmission, display, or control without changing its basic physical nature.
- **Operation:** Encompasses amplification, attenuation, filtering (to remove noise), modulation/demodulation, analog-to-digital conversion (ADC), and linearizing operations.
- **Example:** An operational instrumentation amplifier boosting low-level millivolt signals from a thermocouple to a standard 0–5 V range.

1.4 Data Transmission Element

- **Role:** Transports the conditioned signal over physical distances from the measurement site to a remote control or processing station.
- **Operation:** Utilizes pneumatic lines, coaxial cables, twisted pairs, optical fibers, or wireless telemetry (e.g., 4–20 mA current loops, RS-485, Modbus).
- **Example:** A 4–20 mA industrial current loop transmitter transmitting flow rates over hundreds of meters without line-resistance-induced voltage drops.

1.5 Data Presentation Element

- **Role:** Communicates the measured value to human observers or records it for monitoring and process control.
- **Operation:** Visual display units, analog dials with pointers, digital LED/LCD displays, chart recorders, circular disk loggers, or storage arrays in a SCADA/DCS architecture.
- **Example:** A digital LCD panel displaying temperature in degrees Celsius or a computerized process dashboard showing pressure trends.

2. Classification and Static Characteristics of Instruments

2.1 Classification of Instruments

Basis of Classification	Types	Description
Operating Mode	Deflection Type	The measurand causes an opposing physical deflection that generates an equal and opposing restoring force (e.g., PMMC galvanometer). Fast response, but loads the system.
	Null Type	A known opposing effect balances the measurand to restore a zero indicator (null condition) (e.g., DC Potentiometer, Wheatstone Bridge). High accuracy; does not load the measured system.
Output Form	Analog	Output varies continuously as a smooth function of the input measurand (e.g., dial pointer gauge).
	Digital	Output changes in discrete, quantized increments (e.g., digital multimeter).
Energy Source	Self-Operated (Passive)	Operates entirely on the energy extracted from the measurand (e.g., mercury-in-glass thermometer, Bourdon gauge).
	Power-Operated (Active)	Requires an auxiliary external power source to operate the transducer circuitry (e.g., LVDT, Hall effect sensor).
Method of Verification	Absolute (Primary)	Gives the magnitude of the quantity directly in terms of physical constants without prior calibration (e.g., Tangent Galvanometer).
	Secondary	Quantities are determined directly by observing output deflection; requires prior calibration against an absolute standard (e.g., voltmeter, thermometer).

2.2 Static Characteristics of Instruments

Static characteristics refer to the criteria applied to measurements when the measurand is constant or varies extremely slowly with time.

1. Accuracy and Inaccuracy (Error)

- **Accuracy:** The degree of closeness with which an instrument reading approaches the true or accepted conventional value of the quantity being measured.
- **Point Accuracy:** Accuracy specified only at one particular point on the scale.
- **Percentage of True Value:** Accuracy specified as a percentage of the actual reading:

$$\text{Accuracy} = \pm \left(\frac{\text{Measured Value} - \text{True Value}}{\text{True Value}} \right) \times 100\%$$

- **Percentage of Full-Scale Deflection (% FSD):** Common in commercial specifications; errors can be significant when measuring at the lower end of the scale.

2. Precision

- The degree of mutual agreement among individual measurements made under identical conditions using the same instrument.
- Precision depends on **repeatability** and **reproducibility**.
- *Note:* High precision does not guarantee high accuracy; systematic offsets can make a precise reading inaccurate.

3. Sensitivity

- The ratio of the change in instrument output (Δq_o) to the change in input measurand (Δq_i) under steady-state conditions.

$$S = \lim_{\Delta q_i \rightarrow 0} \frac{\Delta q_o}{\Delta q_i} = \frac{dq_o}{dq_i}$$

- For linear instruments, sensitivity is constant across the entire operating range (the slope of the static calibration curve).

4. Linearity

- The ability of an instrument to reproduce its input linearly at its output.
- **Independent Linearity:** Maximum deviation of any calibration point from the best-fit straight line, expressed as a percentage of the full-scale reading:

$$\text{Linearity (\%)} = \frac{\Delta q_{o(\max)}}{q_{o(\text{full scale})}} \times 100$$

5. Resolution and Threshold

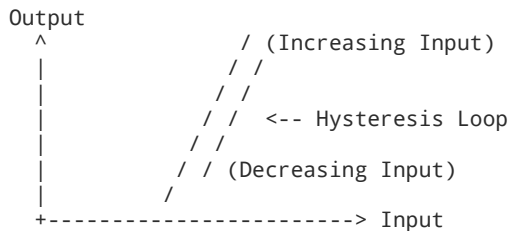
- **Threshold:** The minimum input value required to produce a detectable output change from zero.
- **Resolution:** The smallest incremental change in the input measurand that causes a measurable change in instrument output starting from any non-zero value.

6. Dead Zone and Dead Time

- **Dead Zone:** The largest change in the input value to which the instrument does not respond. Caused by friction, backlash, or play in mechanical linkages.
- **Dead Time:** The time delay between the application of an input change and the first detectable movement of the instrument's output.

7. Hysteresis

- The phenomenon where an instrument gives different output readings for the same input value, depending on whether that value was approached from an increasing or decreasing direction.
- Caused by mechanical friction, magnetic remanence, or internal molecular stress within elastic components.



8. Drift

Gradual degradation or shift of the calibration curve over time, unrelated to changes in the measurand.

- **Zero Drift:** The entire calibration curve shifts parallel up or down due to thermal expansion, component aging, or spring fatigue.
- **Span Drift (Sensitivity Drift):** The slope of the calibration curve changes uniformly across the range.
- **Zonal Drift:** Drift that occurs only within a specific region or fraction of the instrument's operating range.

9. Repeatability vs. Reproducibility

- **Repeatability:** Closeness of agreement between successive measurements of the same measurand carried out under identical conditions: same observer, same location, same instrument, short interval.
- **Reproducibility:** Closeness of agreement under altered conditions: different observers, different laboratories, different environmental conditions, or after extended time intervals.

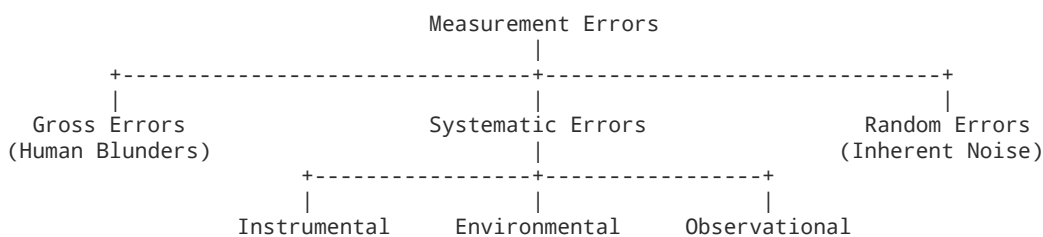
3. Types of Errors and Sources of Error

An error (ϵ) is defined as the algebraic difference between the indicated measured value (A_m) and the true reference value (A_t):

$$\epsilon = A_m - A_t$$

Relative Static Error:

$$\epsilon_r = \frac{A_m - A_t}{A_t}$$



3.1 Types of Errors

1. Gross Errors

- **Nature:** Major discrepancies caused by human faults, misinterpretations, or computational mistakes.
- **Causes:** Misreading scales, incorrect recording of data, improper zero adjustments before use, and improper application of the instrument.
- **Mitigation:** Taking multiple independent readings by different observers; implementing automated data logging.

2. Systematic Errors

These errors are consistent, repeatable, and unidirectional. They obey deterministic physical laws and can be identified, modeled, and corrected.

- **Instrumental Errors:**

- *Inherent Shortcomings*: Imperfections in construction, friction in bearings, weak balance springs, gear backlash, uncalibrated divider resistances.
- *Misuse of Instruments*: Exceeding rated voltages/currents, failure to zero instruments, or using lead wires of incorrect impedance.
- *Loading Effects*: Altering the circuit or system state by connecting the measuring device (e.g., drawing current with a low-impedance voltmeter).

- **Environmental Errors:**

- Caused by external conditions affecting the transducer: temperature shifts, pressure fluctuations, humidity, stray electrostatic or electromagnetic fields, and mechanical vibrations.

- **Observational Errors:**

- Parallax errors from reading pointers at non-perpendicular angles.
- Errors from reading values between analog scale divisions.

3. Random (Accidental) Errors

- **Nature**: Unpredictable, non-deterministic fluctuations remaining after gross and systematic errors are eliminated.
- **Causes**: Thermal agitation noise (Johnson-Nyquist noise), minute mechanical vibrations, friction variations, and line voltage micro-disturbances.
- **Treatment**: Analyzed using statistical and probabilistic methods. Assuming a Gaussian (normal) distribution:

$$P(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- **Best Estimate**: Arithmetic Mean ($\bar{x}$):

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Standard Deviation (σ)**:

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

3.2 Sources of Error (Comprehensive Breakdown)

1. **Improper System Design**: Tolerances in internal electronics (resistor tolerances, op-amp input offset voltages, thermal drift in semiconductor junctions).

2. **Loading Effect**:

- *Electrical*: Connecting an input voltmeter with impedance R_v parallel to circuit load R_L drops the equivalent resistance, changing the measured voltage:

$$V_m = V_{th} \left(\frac{R_v}{R_v + R_{th}} \right)$$

- * *Mechanical*: Accelerometers adding mass to a vibrating cantilever, lowering its natural frequency.
- * *Thermal*: Inserting a heavy thermocouple well into a low-volume liquid stream, cooling the localized area.

3. **Deterioration and Aging**: Magnetic demagnetization of permanent magnets (PMMC meters), oxidation of contact points, loss of elasticity in Bourdon tubes and diaphragms.

4. **Signal Transmission Degradation**: Attenuation, capacitive losses, thermal EMF generation across dissimilar wire junctions, and 50/60 Hz mains inductive pickup.

4. Dynamic Characteristics of Instruments

Dynamic characteristics describe how an instrument behaves when measuring inputs that vary rapidly with time.

4.1 Core Dynamic Terms

- **Speed of Response**: The rapidity with which a measurement system responds to changes in the measured quantity.
- **Measuring Lag**: The delay in an instrument's response to an input change.
 - *Retardation Lag*: The response begins immediately after the input changes, but approaches the final value asymptotically (e.g., a thermometer heating up).
 - *Time-Delay Lag*: The system shows no response for a finite time period following an input change (dead time).

- **Fidelity:** The ability of an instrument to reproduce the shape of an input signal without dynamic distortion.
- **Dynamic Error ($e_d(t)$):** The difference between the true value of the time-varying quantity and the value indicated by the instrument at that moment:

$$e_d(t) = y(t) - x(t)$$

4.2 System Classification by Dynamic Order

The dynamic behavior of generalized linear measurement systems is governed by an n -th order ordinary differential equation:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_1 \frac{dy}{dt} + a_0 y(t) = b_m \frac{d^m x}{dt^m} + \dots + b_0 x(t)$$

Where $x(t)$ is the input measurand and $y(t)$ is the output signal.

1. Zero-Order Systems

- **Equation:** $a_0 y(t) = b_0 x(t) \implies y(t) = K \cdot x(t)$
- **Properties:** No energy storage elements. The output responds instantaneously to the input without distortion or time lag.
- **Dynamic Error:** Zero for any input frequency.
- **Example:** Linear potentiometers used for displacement measurement.

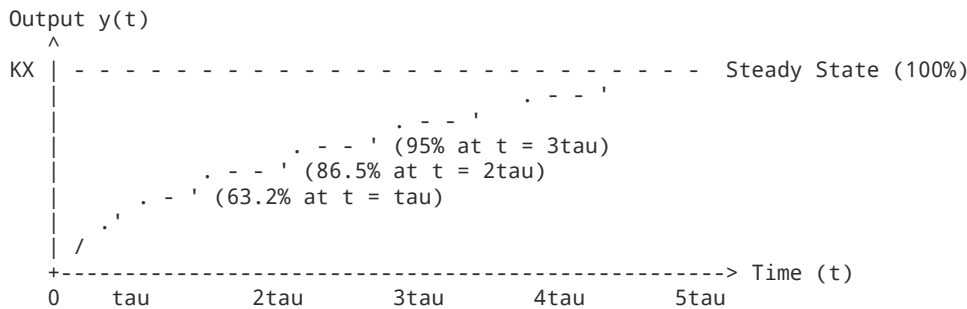
2. First-Order Systems

- Contains a single energy storage element and an energy dissipator.
- **Standard Differential Equation:**

$$\tau \frac{dy(t)}{dt} + y(t) = Kx(t)$$

- Where $\tau = \frac{a_1}{a_0}$ is the **Time Constant** and $K = \frac{b_0}{a_0}$ is the **Static Sensitivity**.
- **Step Response:** For a step input $x(t) = X \cdot u(t)$:

$$y(t) = KX \left(1 - e^{-t/\tau}\right)$$



- At $t = \tau$, $y(\tau) = KX(1 - e^{-1}) \approx 0.632KX$ (the response reaches 63.2% of its final steady-state value).
- The system reaches 98.2% of steady-state value at $t = 4\tau$ and 99.3% at $t = 5\tau$.
- **Frequency Response:** For sinusoidal input $x(t) = X \sin(\omega t)$:

$$M(\omega) = \frac{K}{1 + (\omega\tau)^2}, \quad \phi(\omega) = -\tan^{-1}(\omega\tau)$$

- **Examples:** Bare thermocouple or RTD immersed in a fluid; liquid-in-glass thermometer.

3. Second-Order Systems

- Contains two mutually interactive energy storage elements (e.g., mass-spring-damper, inductive-capacitive circuits).
- **Standard Differential Equation:**

$$\frac{1}{\omega_n^2} \frac{d^2 y(t)}{dt^2} + \frac{2\zeta}{\omega_n} \frac{dy(t)}{dt} + y(t) = Kx(t)$$

- Where $\omega_n = \frac{a_0}{a_2}$ is the **Undamped Natural Angular Frequency** and $\zeta = \frac{a_1}{2a_0a_2}$ is the **Damping Ratio**.

• **Step Response Classifications (for step of height X):**

- **Underdamped** ($\zeta < 1$): Oscillatory response with overshoot before settling:

$$y(t) = KX \left[1 - \frac{e^{-\zeta\omega_n t}}{1 - \zeta^2} \sin(\omega_d t + \phi) \right]$$

where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ and $\phi = \tan^{-1} \left(\frac{1 - \zeta^2}{\zeta} \right)$. * **Critically Damped** ($\zeta = 1$): The fastest response without overshoot:

$$y(t) = KX [1 - (1 + \omega_n t)e^{-\omega_n t}]$$

* **Overdamped** ($\zeta > 1$): Sluggish, non-oscillatory response with a long rise time.

• **Key Transient Metrics (for $\zeta < 1$):**

- **Rise Time** (t_r): Time to rise from 10% to 90% (or 0% to 100%) of final value:

$$t_r = \frac{\pi - \phi}{\omega_d}$$

* **Peak Time** (t_p): Time to reach the maximum peak:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

* **Maximum Percent Overshoot** (M_p):

$$M_p = \exp \left(-\frac{\pi\zeta}{1 - \zeta^2} \right) \times 100\%$$

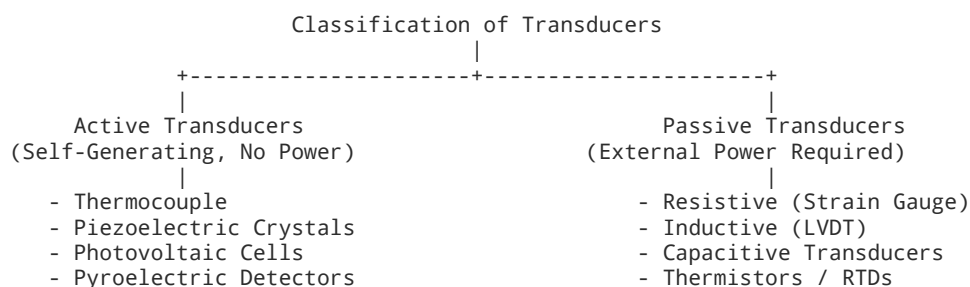
* **Settling Time** (t_s): Time taken to settle within a $\pm 2\%$ error band:

$$t_s \approx \frac{4}{\zeta\omega_n}$$

- **Examples:** Seismic accelerometers, U-tube manometers, PMMC galvanometers.

5. Active and Passive Transducers

A **transducer** converts a physical non-electrical quantity (force, pressure, light, acceleration, displacement) into an electrical output (voltage, current, charge, impedance).



5.1 Comparison Between Active and Passive Transducers

Parameter	Active Transducers	Passive Transducers
Operating Principle	Converts mechanical/thermal/optical energy directly into electrical energy (self-generating).	Modulates an applied external voltage or current by changing its passive electrical parameter (R , L , or C).
External Power	None required. Produces electrical energy directly from the input.	Mandatory. Requires an external DC or AC excitation supply to operate.
Internal Energy Extraction	Extracts a tiny amount of energy directly from the measurand.	Uses external power, though the sensor can still introduce mechanical loading on the measured system.
Equivalent Circuit Model	Voltage or current source (V_{th} or I_N) with internal impedance.	Variable impedance element (R , L , or C).
Circuit Complexity	Simple sensor circuits, but may need high-impedance amplifiers to prevent loading.	Requires bridge networks, oscillators, and demodulation circuits.

Parameter	Active Transducers	Passive Transducers
Representative Examples	Thermocouple (Seebeck), Piezoelectric quartz, Photovoltaic cell.	Strain gauge, Thermistor, LVDT, Variable differential capacitor, RTD.

6. Resistive Transducers

6.1 Principle of Operation

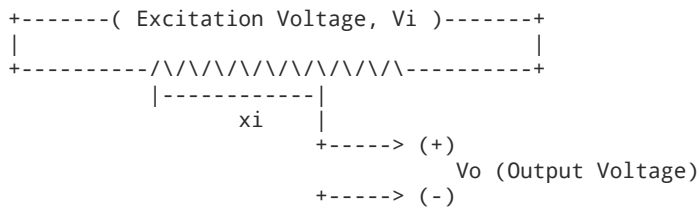
The electrical resistance of a uniform conductor of length L , cross-sectional area A , and specific resistivity ρ is given by:

$$R = \rho \frac{L}{A}$$

Any physical phenomenon that alters L , A , or ρ provides a mechanism for transduction.

6.2 Potentiometric Resistive Transducers

- A resistive track (wire-wound, conductive plastic, cermet, or carbon film) with a sliding wiper connected to the measurand.
- Displacement Mapping:** Translates linear or rotational displacement into a proportional output voltage.

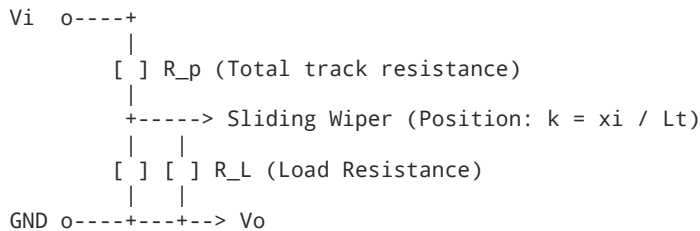


For a total length L_t and wiper displacement x_i :

$$V_o = V_i \left(\frac{x_i}{L_t} \right) \quad (\text{under ideal, unloaded conditions})$$

Loading Effect in Potentiometers

When a voltmeter with finite load resistance R_L is connected across the output, the circuit becomes non-linear:



The output voltage ratio under load is:

$$\frac{V_o}{V_i} = \frac{k}{1 + k(1 - k) \left(\frac{R_p}{R_L} \right)}$$

Where $k = \frac{x_i}{L_t}$ ($0 \leq k \leq 1$) and R_p is total potentiometer resistance.

- If $R_L \rightarrow \infty$, the relationship is strictly linear: $\frac{V_o}{V_i} = k$.
- To minimize the non-linearity error to less than 1%, maintain a high resistance ratio: $\frac{R_L}{R_p} \geq 100$.

7. Thermistors

7.1 Material Composition and Physical Principle

- Definition:** Thermally Sensitive Resistor.
- Composition:** Sintered ceramic mixtures of transition metal oxides (manganese, nickel, cobalt, copper, iron, titanium).
- Operating Mechanism:** As temperature increases, valence electrons are thermally excited into the conduction band, increasing charge carrier density and decreasing electrical resistance. This yields a large **Negative Temperature Coefficient (NTC)**. Positive Temperature Coefficient (PTC) thermistors also exist, made from doped barium titanate ceramics, typically used for overcurrent protection.

7.2 Resistance-Temperature Characteristics

The non-linear relationship between temperature and resistance for an NTC thermistor is given by the exponential relationship:

$$R_T = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$$

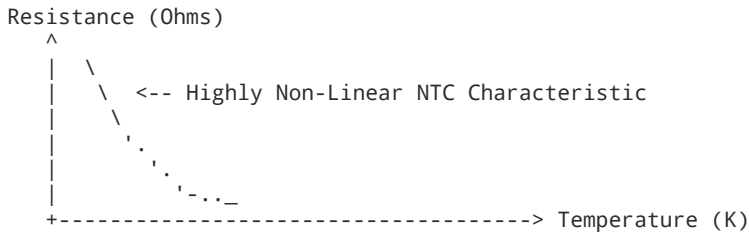
Where:

- R_T = Resistance at absolute temperature T (in Kelvin).
- R_0 = Reference resistance at base temperature T_0 (typically 298.15 K / 25°C).
- β = Material characteristic temperature constant (typically 3000 K to 5000 K).

For wide-temperature, precision calibration, the empirical **Steinhart-Hart Equation** is used:

$$\frac{1}{T} = A + B \ln(R_T) + C [\ln(R_T)]^3$$

Where A , B , and C are calibration coefficients.



7.3 Temperature Coefficient of Resistance (α)

The sensitivity of the thermistor is defined as:

$$\alpha = \frac{1}{R_T} \frac{dR_T}{dT}$$

Differentiating the exponential model gives:

$$\frac{dR_T}{dT} = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right] \cdot \left(-\frac{\beta}{T^2} \right) = -\frac{\beta R_T}{T^2}$$

$$\alpha = -\frac{\beta}{T^2}$$

- At room temperature ($T = 300$ K) with $\beta = 4000$ K:

$$\alpha = -\frac{4000}{(300)^2} \approx -0.0444 \text{ K}^{-1} \Rightarrow -4.44\%/^{\circ}\text{C}$$

- This temperature coefficient is roughly ten times higher than that of platinum RTDs (+0.385%/°C), providing high sensitivity to small temperature changes.

7.4 Thermistor Configurations and Characteristics

- **Constructions:** Tiny beads (glass-encapsulated, diameter 0.15 mm to 1.25 mm), disks, probes, and surface-mount chips.
- **Self-Heating Error:** When current passes through the thermistor, it dissipates power ($P = I^2 R$). If the power is not removed, it raises the internal temperature above that of the ambient environment:

$$\Delta T = \frac{P}{\delta} = \frac{I^2 R_T}{\delta}$$

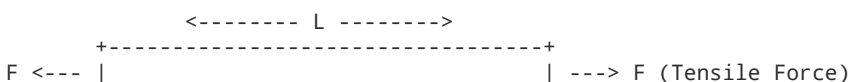
where δ is the **dissipation constant** (mW/°C). To prevent self-heating errors, operating currents should be kept under 100 μA .

8. Strain Gauges

8.1 Piezoresistive Effect and Theoretical Derivation of Gauge Factor

When a metallic conductor is stretched elastically, its resistance changes due to two factors:

1. Physical dimensional changes (elongation along the axis and cross-sectional thinning).
2. Piezoresistivity (changes in electrical resistivity under applied mechanical stress).



+-----+
 <--- A (Cross-section)

Derivation of Gauge Factor (GF)

Starting from the basic resistance formula:

$$R = \rho \frac{L}{A}$$

Take the natural logarithm of both sides:

$$\ln R = \ln \rho + \ln L - \ln A$$

Differentiating both sides:

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dL}{L} - \frac{dA}{A}$$

For a cylindrical wire of diameter D , the area is $A = \frac{\pi}{4} D^2$. Taking logarithms and differentiating yields:

$$\ln A = \ln \left(\frac{\pi}{4} \right) + 2 \ln D \implies \frac{dA}{A} = 2 \frac{dD}{D}$$

Substitute into the fractional resistance change equation:

$$\frac{dR}{R} = \frac{dL}{L} - 2 \frac{dD}{D} + \frac{d\rho}{\rho}$$

Poisson's ratio (ν) relates lateral strain to longitudinal axial strain:

$$\nu = - \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = - \frac{dD/D}{dL/L} \implies \frac{dD}{D} = -\nu \frac{dL}{L}$$

Substitute this relation back into the equation:

$$\frac{dR}{R} = \frac{dL}{L} - 2 \left(-\nu \frac{dL}{L} \right) + \frac{d\rho}{\rho} = \frac{dL}{L} (1 + 2\nu) + \frac{d\rho}{\rho}$$

Divide through by the axial mechanical strain $\epsilon = \frac{dL}{L}$:

$$GF = \frac{dR/R}{\epsilon} = 1 + 2\nu + \frac{d\rho/\rho}{\epsilon}$$

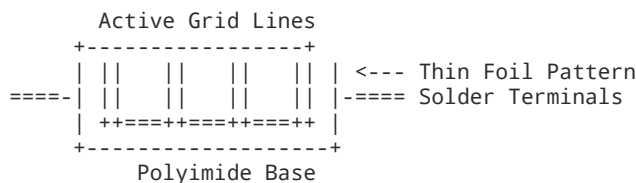
Where:

- 1: Dimensional elongation effect.
- 2ν : Lateral Poisson contraction effect.
- $\frac{d\rho/\rho}{\epsilon}$: **Piezoresistive effect** (crystal lattice spacing shifts affecting electron scattering).

For metallic foil gauges, $1 + 2\nu$ dominates, giving $GF \approx 2.0$ (with $\nu \approx 0.3$). In semiconductor strain gauges, the piezoresistive term dominates, yielding GF values from $+100$ to $+175$ or -50 to -100 .

8.2 Construction and Types of Strain Gauges

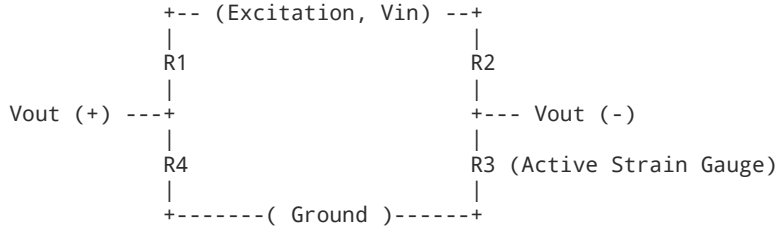
- **Bonded Metallic Foil Gauge:** A thin metal alloy foil (constantan, nichrome, or platinum-tungsten, thickness $3-5 \mu\text{m}$) etched into a zig-zag grid and bonded to a polyimide backing sheet. The grid shape increases the effective length while maintaining a compact footprint.



- **Semiconductor (Piezoresistive) Gauge:** Doped silicon or germanium filaments. They provide high sensitivity, but have a high temperature coefficient of resistance and non-linear response.

8.3 Measurement Bridge Circuits and Temperature Compensation

Because strain variations produce small fractional resistance changes ($\frac{\Delta R}{R} \approx 10^{-4}$ to 10^{-6}), a balanced Wheatstone bridge is required to convert resistance changes into millivolt outputs.



1. Quarter-Bridge Circuit

- $R_1 = R_2 = R_4 = R$, and $R_3 = R + \Delta R$ (single active gauge).
- For an open-circuit output:

$$V_o = V_{in} \left(\frac{R_3}{R_2 + R_3} - \frac{R_4}{R_1 + R_4} \right) = V_{in} \left(\frac{R + \Delta R}{2R + \Delta R} - \frac{1}{2} \right)$$

Assuming $\Delta R \ll 2R$:

$$V_o \approx \frac{V_{in}}{4} \left(\frac{\Delta R}{R} \right) = \frac{V_{in}}{4} (GF \cdot \epsilon)$$

2. Temperature Compensation via Dummy Gauge

- Changes in ambient temperature cause the active gauge resistance to change ($\Delta R_T = R \cdot \alpha \cdot \Delta T$), independent of applied strain.
- To compensate, place an identical, unstrained **dummy gauge** (R_4) on an unloaded section of the same material close to the active gauge (R_3).
- Because R_3 and R_4 are in adjacent arms of the bridge:

$$V_o \propto \left(\frac{\Delta R_3}{R} - \frac{\Delta R_4}{R} \right) = \left(\frac{\Delta R_{\text{strain}} + \Delta R_T}{R} - \frac{\Delta R_T}{R} \right) = \frac{\Delta R_{\text{strain}}}{R}$$

- Thermal expansion and temperature effects cancel out, leaving only the strain measurement.

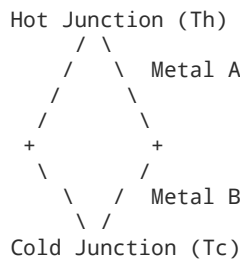
3. Half-Bridge and Full-Bridge Configurations

- **Half Bridge:** Uses two active gauges: one mounted in tension ($R_3 = R + \Delta R$) and one in compression ($R_4 = R - \Delta R$).
 - Sensitivity is doubled: $V_o = \frac{V_{in}}{2} (GF \cdot \epsilon)$, with intrinsic temperature compensation.
- **Full Bridge:** Uses four active gauges (two in tension, two in compression).
 - Sensitivity is quadrupled: $V_o = V_{in} (GF \cdot \epsilon)$, with cancellation of bending or axial cross-axis loads and complete thermal compensation.

9. Thermocouples

9.1 Fundamental Thermoelectric Effects

A thermocouple is an active transducer consisting of two dissimilar electrical conductors joined together at two points, forming a closed loop.



1. Seebeck Effect

When two dissimilar metals A and B are joined at two ends to form junctions held at different temperatures T_h (hot/measuring junction) and T_c (cold/reference junction), an electromotive force (EMF) is generated within the loop:

$$E_{AB} = \int_{T_c}^{T_h} S_{AB}(T) dT = \alpha(T_h - T_c) + \beta(T_h - T_c)^2$$

Where $S_{AB} = S_A - S_B$ is the relative **Seebeck coefficient** (thermoelectric power, in $\mu\text{V}/^\circ\text{C}$).

2. Peltier Effect

The reverse of the Seebeck effect. When an external electric current passes through a junction of two dissimilar conductors, heat is either absorbed or released at the junction depending on current direction, independent of Joule heating ($I^2 R$):

$$\frac{dQ}{dt} = \pi_{AB} \cdot I$$

Where π_{AB} is the **Peltier coefficient**.

3. Thomson Effect

When an electric current flows through an individual homogeneous conductor that has a temperature gradient along its length, heat is absorbed or evolved across the conductor:

$$\frac{dQ}{dt} = \sigma \cdot I \cdot \frac{dT}{dx}$$

Where σ is the **Thomson coefficient**.

The Seebeck effect is the thermodynamic sum of the Peltier and Thomson effects:

$$S_{AB} = \frac{\pi_{AB}}{T}, \quad \frac{dS_{AB}}{dT} = \frac{\sigma_A - \sigma_B}{T} \quad (\text{Kelvin Relations})$$

9.2 Governing Laws of Thermocouples

1. Law of Homogeneous Circuits

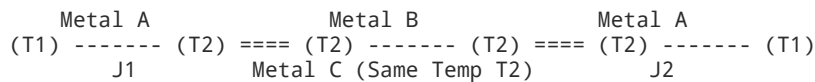
An electric current cannot be sustained in a closed circuit composed of a single, continuous, homogeneous metal by the application of heat alone, regardless of how the cross-section or temperature varies along the path.

- *Practical Implication:* Connecting lead wires will not introduce spurious EMFs as long as the wire material itself is homogeneous.

2. Law of Intermediate Metals

The algebraic sum of the thermoelectric EMFs in a circuit composed of any number of dissimilar metals is zero, provided all junctions are maintained at the same uniform temperature.

$$E_{AC} = E_{AB} + E_{BC}$$



- *Practical Implication:* Instruments (copper leads, digital millivoltmeters, terminal blocks) can be inserted into the circuit without altering the measured EMF, as long as both connection points are kept at the same temperature.

3. Law of Intermediate Temperatures

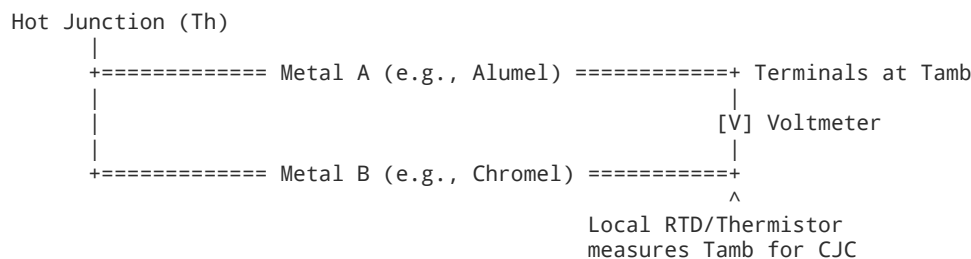
If a thermocouple circuit produces an EMF E_1 with junctions at T_1 and T_2 , and an EMF E_2 with junctions at T_2 and T_3 , the EMF generated when the junctions are at T_1 and T_3 is:

$$E_{T1 \rightarrow T3} = E_{T1 \rightarrow T2} + E_{T2 \rightarrow T3}$$

- *Practical Implication:* Thermocouple calibration charts referenced to 0°C can be corrected for non-zero reference junctions (such as ambient room temperature) without recalibrating the entire probe.

9.3 Cold Junction Compensation (CJC)

Thermocouple voltage tables assume the reference junction (T_{ref}) is held at 0°C (an ice bath). In industrial systems, maintaining an ice bath is impractical.



- **Software/Hardware CJC:**

1. An isothermal block houses the connection terminals between the thermocouple wires and instrument copper traces.

2. A localized semiconductor or RTD temperature sensor measures the block's ambient temperature (T_{amb}).

3. The equivalent voltage for T_{amb} relative to 0°C (V_{CJC}) is calculated using polynomial calibration coefficients:

$$V_{\text{Corrected}} = V_{\text{Measured}} + V(T_{\text{amb}} \rightarrow 0^\circ\text{C})$$

4. The true hot-junction temperature T_h is then derived directly from $V_{\text{Corrected}}$.

9.4 Common Thermocouple Types (ISA Designated)

Type	Positive Conductor	Negative Conductor	Useful Range (°C)	Sensitivity (S at 25°C)	Characteristics & Environment
K	Chromel (90% Ni, 10% Cr)	Alumel (95% Ni, 2% Mn, 2% Al)	−200 to +1260	$\approx 41\ \mu\text{V}/^\circ\text{C}$	Common general-purpose type; works well in oxidizing atmospheres.
J	Iron (Fe)	Constantan (55% Cu, 45% Ni)	−40 to +750	$\approx 51\ \mu\text{V}/^\circ\text{C}$	Low cost; suitable for reducing atmospheres; iron oxidizes rapidly above 760°C.
T	Copper (Cu)	Constantan (Cu-Ni)	−200 to +350	$\approx 40\ \mu\text{V}/^\circ\text{C}$	Resistant to corrosion in moist conditions; useful for cryogenic measurements.
E	Chromel	Constantan	−200 to +900	$\approx 68\ \mu\text{V}/^\circ\text{C}$	Highest output sensitivity among common base-metal types; non-magnetic.
R/S	Platinum - Rhodium (13% / 10% Rh)	Pure Platinum	0 to +1450	$\approx 10\ \mu\text{V}/^\circ\text{C}$	Noble metal construction; high stability and accuracy at elevated temperatures; easily contaminated.
B	Pt - 30% Rhodium	Pt - 6% Rhodium	+600 to +1700	$\approx 10\ \mu\text{V}/^\circ\text{C}$	Used for high-temperature furnace measurements; negligible EMF output below 50°C (CJC often omitted).

10. Differential Output Transducers and LVDT

10.1 Principle of Differential Output Configurations

Differential output transducers use two identical, symmetrical sensing elements connected in electrical opposition.

- **Operating Mechanism:** When an input variable shifts, one sensor's parameter increases ($X_0 + \Delta X$), while the other's decreases by the same amount ($X_0 - \Delta X$).

- **Advantages:**

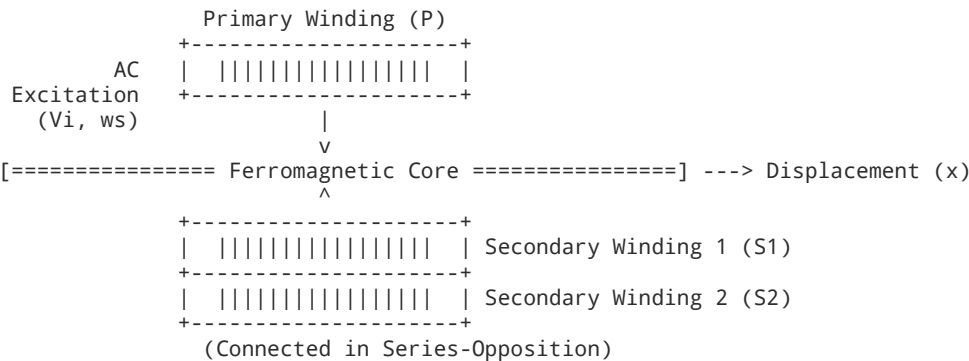
1. **Double Sensitivity:** The output is proportional to $(X_0 + \Delta X) - (X_0 - \Delta X) = 2\Delta X$.
2. **Common-Mode Rejection:** Environmental disturbances (thermal drift, power variations, stray EMI) affect both symmetrical elements identically (X_{noise}), cancelling out in the differential output:

$$V_o \propto (X_1 + X_{\text{noise}}) - (X_2 + X_{\text{noise}}) = X_1 - X_2$$

3. **Improved Linearity:** Odd-order terms are reinforced while even-order harmonic non-linearities are cancelled.

10.2 Linear Variable Differential Transformer (LVDT)

The LVDT is an inductive, passive, differential-output electromechanical transducer that produces an AC voltage output proportional to the mechanical displacement of a movable ferromagnetic core.



1. Construction Details

- **Former:** Hollow, non-magnetic, thermally stable cylindrical ceramic or plastic bobbin.

- **Primary Coil (P):** Wound centrally on the former, excited by an alternating voltage source (typically 1 to 10 V RMS at 50 Hz to 20 kHz).
- **Secondary Coils (S_1 and S_2):** Two identical coils placed symmetrically on either side of the primary coil, wound with equal numbers of turns.
- **Movable Core:** High-permeability, soft iron or nickel-iron alloy cylinder (often slotted axially to minimize eddy current losses), attached to the target object by a non-magnetic rod.
- **Magnetic Shield:** Outer high-permeability enclosure protects the system from stray external magnetic fields.

2. Electrical Operation and Equations

The two secondary coils are connected in **series opposition**, meaning their induced voltages oppose each other:

$$V_{\text{out}} = V_{s1} - V_{s2}$$

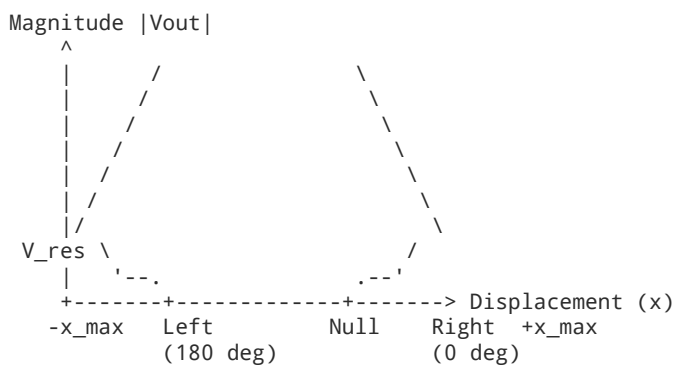
The induced secondary voltages depend on the mutual inductances (M_1 and M_2) between the primary and the respective secondary windings:

$$V_{s1} = M_1 \frac{di_p}{dt}, \quad V_{s2} = M_2 \frac{di_p}{dt}$$

$$V_{\text{out}} = (M_1 - M_2) \frac{di_p}{dt}$$

3. Core Position States

- **Null Position (Zero Displacement, $x = 0$):**
 - The core is centered symmetrically between S_1 and S_2 .
 - Flux coupling is balanced: $M_1 = M_2$.
 - Induced voltages are equal: $V_{s1} = V_{s2}$.
 - Net output ideally cancels to zero: $V_{\text{out}} = 0$ V.
- **Core Displaced Upward / Left ($x > 0$):**
 - Flux linking S_1 increases, while flux linking S_2 decreases: $M_1 > M_2$.
 - $V_{s1} > V_{s2}$.
 - The net output $V_{\text{out}} = (V_{s1} - V_{s2})$ is **in phase** with the primary excitation voltage.
- **Core Displaced Downward / Right ($x < 0$):**
 - Flux linking S_2 increases, while flux linking S_1 decreases: $M_2 > M_1$.
 - $V_{s2} > V_{s1}$.
 - The net output V_{out} undergoes a 180° **phase shift** relative to the primary excitation.



4. Residual Voltage at Null

In practical LVDTs, the output voltage at the null position does not reach zero. A small **residual voltage** remains, typically 0.5% to 1% of full-scale output.

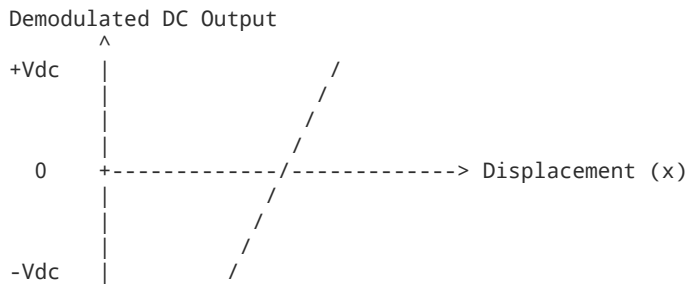
- **Causes:**
 - *Harmonics:* Stray non-linear magnetization curves generate higher-order harmonics (especially 3rd harmonic) in the excitation current that do not cancel out.

- *Phase Shift Differences*: Minor asymmetries in coil winding capacitance and leakage inductances introduce small phase shifts between V_{s1} and V_{s2} .
- *Stray Capacitance*: Asymmetric capacitive coupling between the primary and individual secondary coils.

5. Phase-Sensitive Demodulation (PSD)

Because an AC voltmeter measures only the magnitude $|V_{out}| = |V_{s1} - V_{s2}|$, it cannot determine whether the core moved left or right.

- **Phase-Sensitive Demodulator (PSD)**: Compares the secondary output signal with the original primary AC reference.
- The circuit rectifies the signal synchronously:
 - In-phase signals produce a positive DC voltage ($+V_{dc}$).
 - 180° out-of-phase signals produce a negative DC voltage ($-V_{dc}$).
- The resulting output is a bipolar, linear DC voltage across the measurement range.



6. Operating Characteristics of LVDT

- **Linear Operating Range**: Typically ranges from ± 0.1 mm to over ± 500 mm.
 - **Sensitivity**: Expressed as millivolts output per volt excitation per millimeter displacement (mV/V/mm). Typical ranges: 10 to 100 mV/V/mm .
 - **Frictionless Measurement**: Because the moving core does not make mechanical contact with the coils, there is no mechanical friction or wear. This gives the sensor long operational life and infinite theoretical resolution (limited only by electronic noise).
 - **Dynamic Response**: High dynamic response, limited primarily by the mass of the moving core and the carrier frequency of the AC excitation (which should be at least 10 times the highest vibration frequency measured).
 - **Isolation**: Complete galvanic and physical isolation between the core and the coils.
 - **Limitations**: Sensitive to stray magnetic fields (requires external magnetic shielding); performance can drift with large temperature swings, which alter coil resistance and excitation frequency.
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